GIFT (Green Insights & Financial Transition Platform)

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MegathonX (IIIT-Hyd) — Lloyds Problem Statement 1

recommendation in simple language. It also compares the customer's current financial situation with the suggested green option so the customer can understand both the possible money impact and pollution impact before taking action.

Core idea: Understand the customer → classify transactions → find green opportunities → estimate financial and environmental impact → explain why → show the better possible path.

2. Three User Dashboards

Customer	Bank Officer	Lloyd Employee
Own profile and personalised green recommendations.	Customers using the officer's bank.	All participating banks, officers and their customers.

3. Customer Flow

• **Sign-up — Interface 1:** Full name, phone number, email, password and re-enter password.

• **Sign-up — Interface 2:** Bank name and phone number, with an **Add bank account** option so a customer can add another bank.

• **Account storage:** Store the customer's sign-up details and link the full name, phone number and password for later login.

• **Login:** Full name + phone number + password must match the stored credentials. Otherwise show **Incorrect credentials**.

• **Customer home:** Show the user's details, transaction categories, green opportunities, sustainability impact and explanations.

4. Main Customer Features

• **Transaction Category:** Analyse the customer's expenditure and place transactions into the required sustainability categories.

• **Opportunity Detection:** Identify suitable green opportunities and display them in priority order.

• **Sustainability Impact:** Estimate and display the environmental impact of the current behaviour and the suggested green path.

• **Simple Explanation:** Explain why the recommendation was made using the customer's actual evidence.

• **Financial Case Study:** Compare how much the customer may spend if they continue the current approach versus the suggested green approach, and explain which is financially better under the stated assumptions.

• **Pollution Comparison:** Show estimated pollution from the current path and estimated pollution after following the suggestion.

5. Bank Officer Flow

The bank officer has a login-only entry. The dataset provides the bank ID and its password. The system stores the officer's full name, phone number, bank name, bank ID and password linked to the bank ID. After successful login, the officer can see only customers whose dataset shows usage of the same bank. The officer can view their complete available customer details and the same green-analysis features available for those customers.

MegathonX — Proposed AI Pipeline, Architecture and Testing

6. Lloyd Employee Flow

The Lloyd employee has a login-only entry using the Lloyd ID and password provided in the dataset. The system stores the full name, phone number, Lloyd ID and password linked to the Lloyd ID. After login, the employee can see all bank officers using the platform, regardless of their bank. A bank table shows participating bank names. Selecting a bank opens its customer list; selecting a customer opens the customer's complete available details.

7. High-Level AI / Data Pipeline

Step	Stage	High-level purpose
1	Dataset + customer account data	Customer, bank, transaction and product information are loaded and linked.
2	Customer identity and bank linking	Use phone number, bank details and IDs to connect the correct records and control dashboard access.
3	Transaction analysis	Read the customer's transaction history and classify spending into the required sustainability categories.
4	Customer 360	Combine the customer's transactions, products, banking behaviour and other available signals into one profile.
5	Opportunity prediction	ML estimates propensity for EV Finance, Green Home, Solar Finance and Paperless Banking; confidence is also reported.
6	Eligibility / rules check	Product eligibility follows authorised product rules; a signal or propensity alone does not mean eligibility.
7	Carbon / impact engine	Use documented emissions factors, formulas, baselines and assumptions to estimate potential environmental impact.
8	Financial comparison	Create a simple current-vs-suggested scenario using the available financial data and clearly stated assumptions.
9	Recommendation + explanation	Rank suitable opportunities, show evidence/reason codes and let the LLM explain structured results in simple language.
10	Dashboards	Show personalised customer results, bank-level customer access and Lloyd-level bank/officer/customer views.

8. Proposed System Architecture

Data Layer: customer, bank, officer, product, transaction, vehicle/energy/digital data → **Preprocessing & Feature Layer:** cleaning, linking, transaction categories and customer features → **AI Layer:** transaction classification + propensity models → **Rules Layer:** product eligibility and access control → **Impact Layer:** deterministic carbon calculation + financial scenario comparison → **Recommendation Layer:** priority ranking + reason codes → **LLM Layer:** simple explanation from structured outputs only → **Presentation Layer:** Customer / Bank Officer / Lloyd dashboards + API.

9. Rough Testing Mechanism

• **Login testing:** correct credentials should open the correct dashboard; wrong credentials should be rejected.

• **Access-control testing:** a bank officer must see only customers belonging to the officer's bank; a Lloyd employee can view all participating banks and their officers/customers.

• **Transaction testing:** manually check sample transactions and measure classification correctness and consistency.

• **ML testing:** split customer-level data into training, validation and test sets; compare the model with a simple baseline and report relevant metrics.

• **Recommendation testing:** test normal cases, missing data, conflicting signals and cases where a product should not be recommended.

• **Impact testing:** verify formulas, units, emissions factors, assumptions and repeatability of carbon estimates.

• **Explanation testing:** check that the LLM explanation matches the structured prediction, evidence, impact and product mapping and does not invent facts.

• **End-to-end testing:** trace a customer from raw transactions → prediction → eligibility → impact → recommendation → explanation → dashboard.

10. Responsible AI and Key Principles

The platform will treat all impact values as estimates, document assumptions and data sources, avoid using real confidential banking data, and keep the roles of ML, rules, the carbon engine and the LLM separate. Propensity will not be treated as financial eligibility. The system will also provide evidence and reasons for recommendations so that users can understand why an opportunity was identified.

Planned demo story: Select one customer → show their complete financial/behaviour profile → show categorised spending → show top green opportunities → open “Why?” → show financial and pollution comparison → show estimated impact → show next action. Then switch to Bank Officer and Lloyd views to demonstrate how the same intelligence works at bank and platform level.